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Work/Overseas Experience

Apr., 2026 – Now	RIKEN Special Postdoctoral Researcher (SPDR) in RIKEN AIP Host: Prof. Naoto Yokoya
Oct., 2025 – Mar. 2026	JSPS Postdoctoral Fellow in The University of Tokyo Host: Prof. Naoto Yokoya
Oct., 2024 – Sept., 2025	Research Part-timer in Geoinformatics Team, RIKEN AIP Host: Prof. Naoto Yokoya
Jan., 2024 – July, 2024	Academic Visitor in Photogrammetry and Remote Sensing, ETH Zürich Host: Prof. Konrad Schindler
May, 2023 – Jan., 2024	Research Part-timer in Geoinformatics Team, RIKEN AIP Host: Prof. Naoto Yokoya
Oct., 2022 – Apr., 2023	Research Assistant in Beyond AI Project, The University of Tokyo Host: Prof. Masashi Sugiyama
May, 2021 – Apr., 2022	Trainee in The United Nations Satellite Centre (UNOSAT) Host: Mr. Edoardo Nemni, Mr. Lars Bromley, Dr. Sofia Vallecorsa

Education

Oct., 2022 – Spet., 2025	**Ph.D.** in Complexity Science and Engineering, The University of Tokyo Supervisor: Prof. Naoto Yokoya
Sept., 2019 – June, 2022	**M.E.** in Photogrammetry and Remote Sensing, Wuhan University Supervisor: Prof. Chen Wu, Prof. Bo Du, Prof. Liangpei Zhang IELTS: 7.5, TOEFL: 101, GRE: 333.5
Sept., 2015 – June, 2019	**B.E** in Geomatics Engineering, Anhui University GPA: 94.4/100 (ranking: 1/230)

Publications

G Google Scholar

Book Chapters

[B1] **H. Chen**, J. Song, J. Xia, and N. Yokoya, "Applications in disaster monitoring: From cross-modal 2D mapping to 3D disaster scene understanding," in *Multimodal Remote Sensing Fusion and Classification: Algorithms and Applications*, M.-O. Pun and X. Zhang, Eds., Elsevier, 2026, ch. 8, pp. 247–292, [chapter].

Journal Articles

[J1] Z. Y. Bo, W. Lu, **H. Chen**, S. B. Chen, and B. Luo, "SARU: A shadow-aware and removal unified framework for remote sensing images with new benchmarks," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 239, pp. 39–53, 2026, [paper] [code & dataset].

[J2] **H. Chen**, C. Lan, J. Song, D. Ibañez, J. Xia, K. Schindler, and N. Yokoya, "Multimodal remote sensing change detection: An image matching perspective," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 233, pp. 487–501, 2026, [paper].

[J3] J. Song, **H. Chen**, and N. Yokoya, "Enhancing monocular height estimation via sparse LiDAR-guided correction," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 232, pp. 155–171, 2026, [paper].

[J4] J. Wang, W. Xuan, H. Qi, Z. Chen, **H. Chen**, Z. Zheng, J. Xia, Y. Zhong, and N. Yokoya, "CityVLM: Towards sustainable urban development via multi-view coordinated vision–language model," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 232, pp. 62–74, 2026, [paper].

[J5] Z. Gong, Z. Wei, D. Wang, H. Xiao, X. Ma, **H. Chen**, Y. Jia, Y. Deng, Z. Ji, X. Zhu, X. Yang, N. Yokoya, J. Zhang, B. Du, J. Yan, and L. Zhang, "CrossEarth: Geospatial vision foundation model for domain generalizable remote sensing semantic segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 48, no. 5, pp. 5147–5164, 2026, [paper] [code].

[J6] F. Zhao, F. Xue, J. Wang, J. Yi, D. Zhang, Y. Liu, X. Shao, Y. Chen, J. Song, N. Li, J. Chen, H. Wu, N. Xu, **H. Chen**, and P. Ren, "CoralSMIA: Physically-Grounded Scattering Modeling and Illumination Adaptation for Generalized Underwater Coral Segmentation," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 64, 2026, [paper].

[J7] K. Tang, Z. Zheng, **H. Chen**, X. Chen, and J. Chen, "DreamCD: A change-label-free framework for change detection via a weakly conditional semantic diffusion model in optical VHR imagery," *International Journal of Applied Earth Observation and Geoinformation*, vol. 146, p. 105 125, 2026, [paper] [code].

[J8] Y. Wei, A. Xiao, Y. Ren, Y. Zhu, **H. Chen**, J. Xia, and N. Yokoya, "SARLANG-1M: A benchmark for vision-language modeling in SAR image understanding," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 64, pp. 1–20, 2026, [paper] [dataset].

[J9] L. Wang, Z. Wang, X. Guo, P. Duan, X. Kang, S. Li, Z. Wang, J. Hu, Y. Guo, J. Li, H. Huang, Y. Sheng, Y. Guo, W. He, X. Zeng, Y. Qu, C. Persello, U. Verma, S. Prasad, G. Vivone, **H. Chen**, J. Xia, J. Song, C. Broni-Bediako, K. Kurihara, O. Dietrich, K. Schindler, and N. Yokoya, "All-weather land cover and building damage mapping: Outcome of the 2025 IEEE GRSS data fusion contest," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 19, pp. 21 689–21 707, 2026, [paper].

[J10] X. Shao*, **H. Chen***, F. Zhao, K. Magson, J. Chen, P. Li, J. Wang, and J. Sasaki, "Multi-label classification for multi-temporal, multi-spatial coral reef condition monitoring using vision foundation model with adapter learning," *Marine Pollution Bulletin*, vol. 223, p. 119 054, 2026, [paper].

[J11] **H. Chen**, J. Song, O. Dietrich, C. Broni-Bediako, W. Xuan, J. Wang, X. Shao, Y. Wei, J. Xia, C. Lan, K. Schindler, and N. Yokoya, "Bright: A globally distributed multimodal building damage assessment dataset with very-high-resolution for all-weather disaster response," *Earth System Science Data*, vol. 17, no. 11, pp. 6217–6253, 2025, (ESI hot paper & ESI highly cited paper; Official dataset of 2025 IEEE Data Fusion Contest Track II) [paper] [dataset & code] [challenge].

[J12] J. Xia, **H. Chen**, C. Broni-Bediako, Y. Wei, J. Song, and N. Yokoya, "OpenEarthMap-SAR: A benchmark synthetic aperture radar dataset for global high-resolution land cover mapping [software and data sets]," *IEEE Geoscience and Remote Sensing Magazine*, vol. 13, no. 4, pp. 476–487, 2025, (Official dataset of 2025 IEEE Data Fusion Contest Track I) [paper] [dataset & code] [challenge].

[J13] C. Persello, S. Prasad, U. Verma, G. Vivone, **H. Chen**, J. Xia, J. Song, C. Broni-Bediako, O. Dietrich, K. Schindler, and N. Yokoya, "Report on the 2025 IEEE GRSS data fusion contest: All-weather land cover and building damage mapping [technical committees]," *IEEE Geoscience and Remote Sensing Magazine*, vol. 13, no. 4, pp. 488–492, 2025, [paper].

[J14] Z. Zhao, C. Wu, D. Wang, **H. Chen**, C. Chen, Z. Zheng, B. Du, and L. Zhang, "Advancing weakly-supervised change detection in satellite images via adversarial class prompting," *IEEE Transactions on Image Processing*, vol. 34, pp. 7065–7078, 2025, [paper] [code].

[J15] J. Xu, T. Liu, T. Lei, **H. Chen**, N. Yokoya, Z. Lv, and M. Gong, "CGSL: Commonality graph structure learning for unsupervised multimodal change detection," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 229, pp. 92–106, 2025, [paper] [code].

[J16] C. Persello, S. Prasad, U. Verma, G. Vivone, **H. Chen**, J. Xia, J. Song, C. Broni-Bediako, O. Dietrich, K. Schindler, and N. Yokoya, "2025 IEEE GRSS data fusion contest: All-weather land cover and building damage mapping," *IEEE Geoscience and Remote Sensing Magazine*, vol. 13, no. 2, pp. 388–392, 2025, [paper].

[J17] L. Ding, D. Hong, M. Zhao, **H. Chen**, C. Li, J. Deng, N. Yokoya, L. Bruzzone, and J. Chanussot, "A survey of sample-efficient deep learning for change detection in remote sensing: Tasks, strategies, and challenges," *IEEE Geoscience and Remote Sensing Magazine*, vol. 13, no. 3, pp. 164–189, 2025, (ESI highly cited paper) [paper].

[J18] Z. Zhao, C. Wu, L. Ru, D. Wang, **H. Chen**, and C. Chen, "Plug-and-play disep: Separating dense instances for scene-to-pixel weakly-supervised change detection in high-resolution remote sensing images," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 220, pp. 770–782, 2025, [paper] [code].

[J19] D. Wang, M. Hu, Y. Jin, Y. Miao, J. Yang, Y. Xu, X. Qin, J. Ma, L. Sun, C. Li, C. Fu, **H. Chen**, C. Han, N. Yokoya, J. Zhang, M. Xu, L. Liu, L. Zhang, C. Wu, B. Du, D. Tao, and L. Zhang, "HyperSIGMA: Hyperspectral intelligence comprehension foundation model," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 8, pp. 6427–6444, 2025, (ESI hot paper & ESI highly cited paper) [paper] [code].

[J20] **H. Chen***, J. Song*, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–20, 2024, (ESI hot paper & ESI highly cited paper) [paper] [code].

[J21] **H. Chen**, C. Lan, J. Song, C. Broni-Bedaiko, J. Xia, and N. Yokoya, "ObjFormer: Learning land-cover changes from paired OSM data and optical high-resolution imagery via object-guided transformer," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–22, 2024, (ESI highly cited paper) [paper] [code] [dataset].

[J22] C. Wu, L. Zhang, B. Du, **H. Chen**, J. Wang, and H. Zhong, "UNet-like remote sensing change detection: A review of current models and research directions," *IEEE Geoscience and Remote Sensing Magazine*, vol. 12, no. 4, pp. 2–31, 2024, [paper].

[J23] C. Han, C. Wu, M. Hu, J. Li, and **H. Chen**, "C2F-SemiCD: A coarse-to-fine semi-supervised change detection method based on consistency regularization in high-resolution remote sensing images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–21, 2024, (ESI highly cited paper) [paper] [code].

[J24] C. Broni-Bedaiko, J. Xia, J. Song, **H. Chen**, and N. Yokoya, "Generalized few-shot semantic segmentation in remote sensing: Challenge and benchmark," *IEEE Geoscience and Remote Sensing Letters*, vol. 21, pp. 1–5, 2024, [paper] [code] [dataset] [challenge].

[J25] X. Shao, **H. Chen**, K. Magson, J. Wang, J. Song, J. Chen, and J. Sasaki, "Deep learning for multilabel classification of coral reef conditions in the indo-pacific using underwater photo transect method," *Aquatic Conservation: Marine and Freshwater Ecosystems*, vol. 34, no. 9, e4241, 2024, (Front cover paper; Journal's most-cited papers) [paper] [dataset].

[J26] **H. Chen**, J. Song, C. Wu, B. Du, and N. Yokoya, "Exchange means change: An unsupervised single-temporal change detection framework based on intra- and inter-image patch exchange," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 206, pp. 87–206, 2023, [paper] [code].

[J27] **H. Chen**, N. Yokoya, and M. Chini, "Fourier domain structural relationship analysis for unsupervised multimodal change detection," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 198, pp. 99–114, 2023, (ESI highly cited paper) [paper].

[J28] C. Han, C. Wu, H. Guo, M. Hu, J. Li, and **H. Chen**, "Change guiding network: Incorporating change prior to guide change detection in remote sensing imagery," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 16, pp. 8395–8407, 2023, (ESI highly cited paper) [paper] [code].

[J29] C. Han, C. Wu, H. Guo, M. Hu, and **H. Chen**, "HANet: A hierarchical attention network for change detection with bitemporal very-high-resolution remote sensing images," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 16, pp. 3867–3878, 2023, (ESI highly cited paper) [paper] [code].

[J30] **H. Chen**, N. Yokoya, C. Wu, and B. Du, "Unsupervised multimodal change detection based on structural relationship graph representational learning," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–18, 2022, [paper] [code].

[J31] C. Wu, **H. Chen**, B. Du, and L. Zhang, "Unsupervised change detection in multitemporal VHR images based on deep kernel PCA convolutional mapping network," *IEEE Transactions on Cybernetics*, vol. 52, no. 11, pp. 12 084–12 098, 2022, [paper] [code].

[J32] C. Wu, Y. Guo, H. Guo, J. Yuan, L. Ru, **H. Chen**, B. Du, and L. Zhang, "An investigation of traffic density changes inside Wuhan during the COVID-19 epidemic with GF-2 time-series images," *International Journal of Applied Earth Observation and Geoinformation*, vol. 103, p. 102 503, 2021, [paper].

[J33] **H. Chen**, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep siamese convolutional multiple-layers recurrent neural network," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 4, pp. 2848–2864, 2020, (ESI highly cited paper) [paper] [code].

Conference Proceedings

[C1] Y. Wei, A. Xiao, **H. Chen**, J. Xia, and N. Yokoya, "MM-OVSeg: Multimodal optical–SAR fusion for open-vocabulary segmentation in remote sensing," in *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, [code] [dataset].

[C2] Z. Zhao, C. Wu, X. Cao, D. Wang, **H. Chen**, D. Tang, L. Zhang, and Z. Zheng, "Change-Bridge: Spatiotemporal image generation with multimodal controls for remote senisng," in *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, [paper].

- [C3] A. Xiao, S. Cheng, Y. Xu, Y. Ren, **H. Chen**, and N. Yokoya, "GeoMMBench and GeoMMAgent: Toward expert-level multimodal intelligence in geoscience and remote sensing," in *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, (Highlight) [paper] [code].
- [C4] J. Wang, W. Xuan, H. Qi, Z. Liu, K. Liu, Y. Wu, **H. Chen**, J. Song, J. Xia, Z. Zheng, and N. Yokoya, "DisasterM3: A remote sensing vision-language dataset for disaster damage assessment and response," in *Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025, [paper] [code] [dataset].
- [C5] J. Song, **H. Chen**, W. Xuan, J. Xia, and N. Yokoya, "SynRS3D: A synthetic dataset for global 3D semantic understanding from monocular remote sensing imagery," in *Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024, (Spotlight) [paper] [code] [dataset].
- [C6] **H. Chen**, J. Song, and N. Yokoya, "Change detection between optical remote sensing imagery and map data via segment anything model (SAM)," in *Proceeding of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024, pp. 8509–8512, (Oral) [paper].
- [C7] N. Yokoya, J. Xia, J. Song, C. Broni-Bedaiko, and **H. Chen**, "OpenEarthMap benchmark suite and its applications," in *Proceeding of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024, pp. 959–962, (Oral) [paper].
- [C8] J. Song, **H. Chen**, and N. Yokoya, "SyntheWorld: A large-scale synthetic dataset for land cover mapping and building change detection," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024, pp. 8287–8296, (Poster) [paper] [dataset].
- [C9] **H. Chen**, E. Nemni, S. Vallecorsa, X. Li, C. Wu, and L. Bromley, "Dual-tasks siamese transformer framework for building damage assessment," in *Proceeding of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2022, pp. 1600–1603, (Oral) [paper].
- [C10] **H. Chen**, C. Wu, B. Du, and L. Zhang, "Deep siamese multi-scale convolutional network for change detection in multi-temporal VHR images," in *10th International Workshop on the Analysis of Multitemporal Remote Sensing Images*, 2019, pp. 1–4, (Oral) [paper].

Fundings

July, 2026	Grant-in-Aid Research Activity Start-up (**PI**, 2,600,000 JPY) [link]
Apr., 2026	RIKEN SPDR Research Fund (**PI**, 3,500,000 JPY) [link]
Sept., 2024	UTokyo AI Center Fusion Research Promotion Fund (**PI**, 1,600,000 JPY) [link]
Apr., 2024	Grant-in-Aid for JSPS Research Fellows (**PI**, 1,500,000 JPY) [link]
Sept., 2023	UTokyo AI Center Fusion Research Promotion Fund (**PI**, 2,000,000 JPY) [link]
June, 2023	UTokyo GSFS Challenging New Area Doctoral Research Grant (**PI**, 600,000 JPY) [link]
June, 2023	UTokyo GSFS Challenging New Area Doctoral Research Grant (600,000 JPY) [link]
Mar., 2023	Microsoft Research Asia Collaborative Research Grant (**PI**, 10,000 USD)

Awards & Honors

Mar., 2026	UTokyo President's Award for Academic Achievement [link] [news]

Mar., 2026	UTokyo GSFS Dean's Award for Academic Achievement [link]
Apr., 2025	JSPS Research Fellow's Special Allowance (Research Incentive Grant) [link]
Apr., 2024	JSPS Research Fellowships for Young Scientists DC2 (Selection ratio: 17.1%) [link]
Dec., 2023	Japanese-Swiss Young Researchers' Special Exchange Grant [link]
Oct., 2022	The University of Tokyo Fellowship (one of 21 Ph.D. students selected) [link]
June, 2022	Outstanding Graduates of Wuhan University
Dec., 2021	Wang Zhizhuo Innovation Talent Scholarship (Top 1‰) [link]
Oct., 2021	National Scholarship for Postgraduates (Top 1%) [link]
Oct., 2020	National Scholarship for Postgraduates (Top 1%) [link]
Oct., 2020	First Prize of Wuhan University Scholarship for Excellent Postgraduate (Top 5%)
Sept., 2019	LIESMARS Scholarship for Excellent First-Year Postgraduates (Top 9 in 169)
May, 2019	Excellent Graduate of Anhui Province, China (Top 1%)
Nov., 2018	Second Prize of Esri Cup GIS Software Development Contest in China (Top 5%)
Oct., 2018	First Prizes of Academic Scholarship of Anhui University (Top 3%)
July, 2018	Outstanding Prize of National Geomatics Contest in Programming (Top 2%)
Apr., 2018	Meritorious Winner of the U.S. Mathematical Contest in Modeling
Nov., 2017	Second Prize of China National Mathematical Contest in Modeling (Top 5%)
Oct., 2017	National Scholarship for Undergraduates (Top 1%)

Academic Service

Community Service

2025 – Now	Co-Lead of Machine/Deep Learning for Image Analysis (MIA) working group in IEEE GRSS Image Analysis and Data Fusion Technical Committee (IADF TC)
2026	Judge for the Outstanding Student and PhD candidate Presentation (OSPP) contest of The EGU General Assembly 2026
2026	Lead-Organizer of Bright Challenge: Advancing All-Weather Building Damage Mapping to Instance-Level in CVPR Workshop 2026
2025	Lead-Organizer of IEEE GRSS Data Fusion Contest 2025 – All-Weather Land-Cover Mapping & Disaster Response
2024	Co-Organizer of OpenEarthMap Land Cover Mapping Few-Shot Challenge in CVPR Workshop 2024

Editorial Activity

2026	Session Chair on Synthetic Data for Earth Observation: Bridging the Data Gap to Empower Next-Generation Remote Sensing Applications in *IGARSS 2026*
2026	Special Issue on Spatio-temporal Intelligence Foundation Models and Applications for intelligent Earth Observation in *Geo-Spatial Information Science*
2024 – 2026	Special Issue on Recent Advances in Deep Learning-based High-Resolution Image Processing and Analysis, Remote Sensing for Building Extraction and Large-Scale Urban Mapping in *Remote Sensing*

Reviewer

Journal	*IEEE TPAMI, IEEE TIP, IEEE TNNLS, IEEE TCSVG, IEEE TMM, IEEE TGRS, IEEE GRSL, IEEE JSTARS, IJCV, RSE, ESSD, ISPRS J P&RS, MedIA, JAG, PR, Information Fusion, Artificial Intelligence Review, GIScience & Remote Sensing, EAAI, NN, GSIS, IJDE, ESWA, Advanced Engineering Informatics, Neurocomputing, Ecological Indicators, Ecological Informatics, Remote Sensing Applications: Society and Environment, npj Heritage Science, and other 30 journals...*
Conference	*CVPR, ECCV, NeurIPS, AAAI, ACMMM, IGARSS and other conferences...*

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